COMP1832	**Programming Fundamentals forData Science**	**Faculty HeaderID:**	**Contribution 100% of course**
Course Leader **Dr. Jia Wang**	**COMP1832 Portfolio**		**Deadline Date: 08/12/2025**

This coursework should take an average student who is up-to-date with tutorial work approximately 30 hours

Feedback and grades are normally made available within 15 working days of the coursework deadline

Learning Outcomes:

1. Demonstrate knowledge and understanding of commonly used data structures and processing techniques for Data Science.
2. Understand and implement processing pipelines for use in Data Science applications.
3. Build efficient solutions for data manipulation.

Plagiarism is presenting somebody else's work as your own. It includes: copying information directly from the Web or books without referencing the material; submitting joint coursework as an individual effort; copying another student's coursework; stealing coursework from another student and submitting it as your own work. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set down by the University. Please see your student handbook for further details of what is / isn't plagiarism.

All material copied or amended from any source (e.g. internet, books) must be referenced correctly according to the reference style you are using.

Your work will be submitted for plagiarism checking. Any attempt to bypass our plagiarism detection systems will be treated as a severe Assessment Offence.

Aims

This module provides a hands-on approach to the programming and applied mathematics expertise that is essential to the Data Science programme. Students will develop their understanding and gain practical experience with the data structures and processing algorithms most frequently used in developing Data Science solutions. The module is complementary to the other topics covered in the master's programme and will prepare students in their learning experience as well as in their further pursuit for higher education or work in industry.

Indicative content

This module introduces the basic programming and mathematics knowledge required for successfully building Data Science applications. A range of the following topics will be covered including, but not limited to:

- Basic maths concepts and their computer implementation: Vectors, Matrices, Multi-dimensional Arrays
- Programming data processing tools with Python
- Programming data processing tools with R
- Statistics in data science
- Data visualisation techniques

Learning and Teaching Activities

A combination of weekly 1-hour lectures/tutorials (33%) and 2-hour lab sessions (67%).

This course is to be delivered via several complementary activities: lectures/tutorials, practical work and directed unsupervised learning. The rationale for this mix of activities is to give the students an interesting and varied learning experience combining theory and analysis to underpin the core practical work.

Students will also have extra support through supplemental material in the form of digital media, tutorials and example projects to analyse and disseminate.

Coursework Submission Requirements

- An electronic copy of your work for this coursework must be fully uploaded on **Monday 08/12/2025 latest at 17:00pm** using the links on the module Moodle page for COMP1832.
- For this coursework you must submit **a single PDF with no more than 3-pages** document which include

solutions to both Python and R portfolio tasks. **A penalty will be imposed if the submission exceeds the specified page limit.**

- **This PDF must include answers to all questions, justifications for the chosen methods and solutions, required plots as specified in the tasks, and interpretations of those plots. Only include code if explicitly required by the task.**
- **A section called 'Declaration of AI Use' at the end of the report.**
- In general, any text in the PDF document must not be an image (i.e. must not be scanned) and would normally be generated from other documents (e.g. MS Office using"Save As .. PDF"). An exception to this is hand written mathematical notation,but when scanning do ensure the file size is not excessive.
- There are limits on the file size (<=500MB).
- Make sure that any files you upload are virus-free and not protected by a password or corrupted otherwise they will be treated as null submissions.
- Your work will not be printed in colour. Please ensure that any pages with colour are acceptable when printed in Black and White.
  - You must NOT submit a paper copy of this coursework.
  - All coursework must be submitted as above. Under no circumstances can they be accepted by academic staff

The University website has details of the current Coursework Regulations, including details of penalties for late submission, procedures for Extenuating Circumstances, and penalties for Assessment Offences.

See http://www2.gre.ac.uk/current-students/regs

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Detailed Coursework Specification

- **Tasks:**

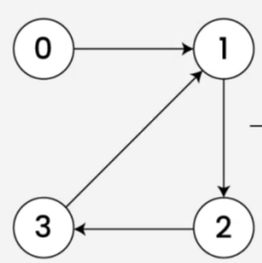
Portfolio (100 Marks)

<<<<<<<<<<<<<<<<<<<<<Python part start>>>>>>>>>>>>>>>>>>>>>>>>>

Examine the provided Python solution (Moodle: Tasks and Dataset→COMP1832_CW_tasks.pdf) that constructs and visualises a segment of the London Underground network using graph and complete the following three tasks.

1. **Based on the code of the graph construction, provide the adjacency matrix showing the connectivity of all the stations (10 mark).**

   An example of an adjacency matrix (right), corresponding to the graph (left), is shown below. The first row and column (0, 1, 2, 3) represent node names, while the remaining entries (0 or 1) indicate connectivity: 0 means no edge; 1 means an edge exists.



→

	0	1	2	3
0	0	1	0	0
1	0	0	1	0
2	0	0	0	1
3	0	1	0	0

2. **Based on the visualisation (Figure 1), identify two aspects of the plot that need be improved and justify why these improvements are necessary from the perspective of using the visualisation for travel using multiple tube lines (10 mark).**

3. **Implement the two improvements suggested in task 2 (20 mark) and generate a new plot (10 mark). Include code, code comments, and plot interpretation in your submission. Re-use the provided code that constructs the graph. You must ONLY make changes that are related to the proposed improvement.**

   Specify what chunk of code needs to be revised using the provided line numbers. For example:

The code from L33 to L35 will be corrected as:
XXXXX (insert the corrected code here).

<<<<<<<<<<<<<<<<<<<<<Python part end>>>>>>>>>>>>>>>>>>>>>>>>>

<<<<<<<<<<<<<<<<<<<<<<<<R part start>>>>>>>>>>>>>>>>>>>>>>>>

Review the R solution (Moodle: Tasks and Dataset→COMP1832_CW_tasks.pdf) provided for the given analytics requirements and answer the following questions.

1. **Figure 2 isn't a correct solution to show the dispersion and central tendency of "Banking and credit industry fraud". What caused the incorrect visualisation?**

   Specify the issue in relation to the provided code (10 mark) and suggest a possible correction in the code (5 mark). Generate a new plot based on the correction (10 mark).

2. **Figure 3 is also wrong. What caused the wrong visualisation?**

   Specify the issue in relation to the provided code (10 mark) and suggest a possible correction in the code (5 mark). Generate a new plot based on the corrections and provide plot interpretations on the fraud regional pattern (10 mark).

For both R tasks, specify what chunk of code needs to be revised using the provided line numbers. For example:

The code from L33 to L35 will be corrected as:
XXXXX (insert the corrected code here).

<<<<<<<<<<<<<<<<<<<<<<<<R part end>>>>>>>>>>>>>>>>>>>>>>>>

Add the section of 'Declaration of AI use' at the end of your PDF.

- **Deliverables:**

An admissible coursework submission needs to include:

⇒ Compile all required solutions, including answers to the questions, revised code, justifications, plots and plot interpretations, into **a single PDF file** with a maximum length of **3 pages**.
⇒ Add the section of 'Declaration of AI use' at the end of your PDF.
⇒ Upload the PDF by Monday, 08/12/2025, no later than 17:00, using the submission link provided on the coursework Moodle page.
⇒ A **10% penalty** will be applied if the submission exceeds the 3-page limit.

Your submission must not include any additional content. Do not include a title page (anonymous marking applies, so do not reveal your name or student ID anywhere in the submission), references, or copies of task instructions and descriptions. Less is more.

- **Grading Criteria**

80%-100% Exceptional	You will need to have: an excellent implementation and reflection on understanding your tasks. All requirements are implemented to a higher standard.
70-79% Excellent	A very good implementation showing your solutions with all requirements implemented. Code and plots are clear and readable with justified descriptions and explanations.
60-69% Very good	A good implementation showing your solutions to all the tasks: all required plots are implemented, and both python and R code are working. Explanations of your

	solutions are provided which reflect good understanding of given data analytics tasks.
50-59% Good	An implementation showing your basic understanding and programming skills of data transformation, basic statistical analyses, and visualisation. Providing solutions with minimum requirements implemented with some justifications.
0-49% Fail	A portfolio with very limited tasks solved. No solution or very few solutions provided for the assignment. A portfolio that fails in reflecting the understanding of the basics of processing and visualising data in Python and R.

A 10% penalty will be applied if your submission exceeds the three-page limit. For example, if your original mark is 65% and your submission contains five pages, the final penalised mark will be 5